Lecture 7 · Uncertainty I: Expected Utility and Risk Aversion

MWG Ch. 6.A–6.C · utility finally goes cardinal

Advanced Microeconomics 1

Thursday 22 October

The two goals, under uncertainty

Same frame, riskier world

(a) Is expected utility the right model of choice under risk? *(Testably, on you.)* **(b)** What is risk reduction worth — and how should insurance, private and social, be designed?

- Everything so far assumed known consequences; almost nothing interesting has them. Alternatives become **lotteries**, and the assumption ladder gains new rungs: rationality (carried over) $\rightarrow$ **independence** buys a representation linear in probabilities $\rightarrow$ **probabilistic sophistication** (Monday) buys subjective probabilities where none are given.
- Each rung has a named breaker — and this room has personally activated all of them: Allais today, your warm-up answers today, your jars on Monday. This is the block where the bill is almost entirely *home-grown*.
- Structural payoff on the way up: with independence, utility finally goes **cardinal** — curvature of u will *mean* something (risk aversion), and that curvature is goal (b)'s currency.

Lotteries and the independence axiom

Lotteries

Outcomes $C = \{c_1, \ldots, c_N\}$ (finite for today).

Definition · simple and compound lotteries (MWG 6.B.1–6.B.2)

A **simple lottery** is $L = (p_1, \ldots, p_N)$, $p_n \geq 0$, $\sum_n p_n = 1$. A **compound lottery** $(L_1, \ldots, L_K; \alpha_1, \ldots, \alpha_K)$ yields simple lottery L_k with probability α_k.

- Maintained hypothesis (*consequentialism*): only the **reduced** simple lottery matters — $\sum_k \alpha_k L_k$. How uncertainty resolves, in how many stages, is irrelevant. Already substantive: rules out anxiety, suspense, preference for one-shot vs. gradual resolution.
- Geometry: the set of simple lotteries is the simplex Δ^{N-1}; with $N = 3$, a triangle — the canvas for everything today.

The axioms

Preferences $\succsim$ on the space of simple lotteries $\mathcal{L}$, assumed rational, plus:

Definition · continuity (MWG 6.B.3)

For all L, L', L'': the sets $\{\alpha \in [0,1] : \alpha L + (1-\alpha)L' \succsim L''\}$ and $\{\alpha : L'' \succsim \alpha L + (1-\alpha)L'\}$ are closed. (No lexicographic-style jumps; small probability changes don't reverse strict rankings.)

Definition · independence (MWG 6.B.4)

For all L, L', L'' and $\alpha \in (0,1)$:

$$L \succsim L' \quad \Longleftrightarrow \quad \alpha L + (1-\alpha)L'' \succsim \alpha L' + (1-\alpha)L''.$$

- Mixing both sides with a common third lottery cannot reverse a ranking: the L''-branch happens or it doesn't, and if it doesn't you face the original comparison. No analogue exists in consumer theory — there, mixing commodity bundles creates genuinely new bundles; here, mixing lotteries only reshuffles probability

Theorem · v.N–M representation (MWG 6.B.2–6.B.3)

$\succsim$ on $\mathcal{L}$ is rational, continuous, and satisfies independence **iff** there exist numbers $u_1, \ldots, u_N$ such that

$$L \succsim L' \iff \sum_n p_n u_n \geq \sum_n p'_n u_n.$$

Moreover $\tilde{U}$ represents the same $\succsim$ in expected-utility form **iff** $\tilde{u}_n = \beta u_n + \gamma$, $\beta > 0$: unique up to positive affine transformations.

- Linearity in probabilities is the whole content. In the simplex: indifference curves are **parallel straight lines**.
- The uniqueness clause is the cardinality: ratios of utility *differences* are meaningful. "The gain from c_1 over c_2 is twice that from c_2 over c_3" now says something — it predicts choices among lotteries.

Proof sketch: the calibration construction

Assume best and worst outcomes $\bar{L} \succ \underline{L}$ (else trivial). Three steps:

1. **Monotonicity in the mixing weight**: $\beta\bar{L} + (1-\beta)\underline{L} \succ \gamma\bar{L} + (1-\gamma)\underline{L}$ iff $\beta > \gamma$ — from independence.
2. **Calibrate**: by continuity, for each lottery L there is a unique $\alpha_L \in [0,1]$ with

$$L \sim \alpha_L \bar{L} + (1-\alpha_L)\underline{L}. \qquad \text{Define } U(L) = \alpha_L.$$

3. **Linearity**: independence forces $\alpha_{\beta L+(1-\beta)L'} = \beta\alpha_L + (1-\beta)\alpha_{L'}$ — substitute each lottery by its calibrated equivalent inside the mixture. □ (sketch)

- Compare Lecture 2's Debreu construction: measure a bundle by *where its indifference curve crosses the diagonal*; measure a lottery by *its indifference mixture of best and worst*. Same trick, new clothes — and this one delivers cardinality because mixing is linear.

Allais: independence meets the lab

Choose in each pair (amounts in millions):

Pair 1:	L_1: \$1 for sure	vs.	L_1': \$5 w.p. .10,	\$1 w.p. .89,	\$0 w.p. .01
Pair 2:	L_2: \$1 w.p. .11, \$0 w.p. .89	vs.	L_2': \$5 w.p. .10,	\$0 w.p. .90	

- Modal pattern: $L_1 \succ L_1'$ *and* $L_2' \succ L_2$. Subtract the common consequence (\$1 w.p. .89 vs. \$0 w.p. .89) and independence says the two pairs are *the same choice*. The modal pattern violates it.
- In the simplex: indifference lines that fan out instead of staying parallel. Responses: certainty effects, rank-dependent weighting, prospect theory (pointers Monday).
- MWG's own verdict is worth quoting in spirit: a theory this useful is not discarded for failing a boundary experiment — but you must know *where* the boundary is.

Money lotteries and risk aversion

Risk aversion = concavity

Outcomes now amounts of money; lottery = distribution F; Bernoulli utility $u(x)$, continuous and increasing; $U(F) = \int u(x)\, dF(x)$.

Definition · risk aversion (MWG 6.C.1)

The agent is **risk averse** if for every F: $u\left(\int x\, dF\right) \geq \int u(x)\, dF$ — the expected value for sure beats the lottery. By Jensen's inequality, this holds **iff u is concave**.

Two equivalent measurements:

- **Certainty equivalent** $c(F, u)$: $u(c(F, u)) = \int u\, dF$; risk aversion $\iff c(F, u) \leq \int x\, dF$.
- **Risk premium** $\pi = \int x\, dF - c(F, u) \geq 0$: what the agent pays to shed the risk — the insurance industry's revenue, in one symbol.

Arrow–Pratt: how curved, in usable units

goal (b)'s currency

Definition · coefficients of risk aversion (MWG 6.C.2)

$$r_A(x) = -\frac{u''(x)}{u'(x)} \quad \text{(absolute)}, \qquad r_R(x) = -\frac{x\,u''(x)}{u'(x)} \quad \text{(relative)}.$$

- Why this ratio: u'' alone is meaningless (affine transformations rescale it); dividing by u' makes r_A invariant — the *first* cardinal statistic of the course.
- Where it earns its units: for a small risk ε with mean 0 and variance σ^2 around x,

  $$\pi \;\approx\; \tfrac{1}{2}\,\sigma^2\, r_A(x)$$

  (second-order Taylor of both sides of the CE definition — a two-line derivation you must do cold).
- **CARA**: $u = -e^{-ax}$, $r_A \equiv a$ (wealth-invariant absolute). **CRRA**: $u = x^{1-\sigma}/(1-\sigma)$, $r_R \equiv \sigma$ (wealth-invariant relative — why growth theory scales, and why every macro model is CRRA).

Comparing people, comparing wealth levels

Proposition · Pratt (1964); MWG 6.C.2

The following are equivalent: (i) $r_A^2(x) \geq r_A^1(x)$ for all x; (ii) $u_2 = \psi(u_1)$ for some increasing concave ψ; (iii) $c(F, u_2) \leq c(F, u_1)$ for every F; (iv) whenever agent 2 accepts a gamble, so does agent 1.

- One local statistic (r_A) controls a global behavioral ordering — the reason Arrow–Pratt is *the* language of applied risk work.
- Wealth effects: under **decreasing** absolute risk aversion (DARA — the empirically sane case), richer agents accept more absolute risk; the risky asset is a normal good in the classic portfolio problem (MWG 6.C.4–6.C.5, EC3).

How risk averse are people? Your data, and the field's

Two questions you answered in September

In the Session 1 quiz, before any of this theory:

Q1	50–50: lose €100 / gain €110. Accept?
Q2	50–50: lose €1,000 / gain €10,000,000. Accept?

- Your answers: [insert: *n* rejected Q1; of those, *m* accepted Q2].
- Hold that pattern on screen. Under expected utility, it is about to become a problem.

Rabin's calibration theorem

Theorem · Rabin (Econometrica 2000), flavor

Suppose an expected-utility maximizer (any increasing concave u) rejects the 50–50 lose \$100 / gain \$110 gamble *at every wealth level*. Then they reject a 50–50 lose \$1,000 / gain *any sum* gamble.

- Mechanism, in one breath: rejecting the small bet at w bounds $u'(w+110) \leq \frac{100}{110} u'(w-100)$; rejecting it *everywhere* compounds that decay geometrically, so marginal utility collapses over a few thousand dollars — and no finite prize can compensate a \$1,000 loss.
- So EU cannot explain both of: meaningful small-stakes risk aversion *and* sane large-stakes behavior. Concavity of u over wealth is a theory of *large* risks only.
- Your class pattern — reject small, accept large — is not a preference an EU maximizer can have. Candidate diagnoses: **loss aversion** and **narrow bracketing** (Rabin–Thaler 2001): the small gamble is evaluated in isolation, gains vs. losses, not merged into lifetime wealth.

In passing: the field agrees with the lab

- Sydnor (2010): homeowners pay real premia to shave a few hundred dollars off deductibles against few-percent claim probabilities — Rabin's theorem with a price tag.
- Chetty (AER 2006): labor-supply responses bound the curvature of consumption utility — *modest*, low single digits, far below what small-stakes choices imply. The tension is the point.
- Working rule for a career: CRRA σ in the low single digits for large risks; do not ask concave-u-over-wealth to explain lottery tickets or phone insurance.

Scorecard, takeaways, readings

Takeaways

- **Goal (a):** hypothesis bought (independence $\Rightarrow$ EU, linear in probabilities, cardinal at last) — and *twice cracked*, once by Allais, once by this room's own warm-ups via Rabin's calibration. EU is a theory of *large* risks.
- **Goal (b):** its currency is minted — certainty equivalents, risk premia, $\pi \approx \frac{1}{2}\sigma^2 r_A$, Pratt's global ordering. Monday spends it: the insurance theorem, and an optimal social-insurance benefit computed from two estimable statistics.

Required: MWG 6.A–6.C. **Next:** 6.D–6.F for Monday.
The block's papers: *(a)* Choi–Fisman–Gale–Kariv (2007) — your 25 rounds are its design; skim before Monday. *(b)* Chetty (JPubE 2006), the Baily–Chetty formula — intro before Monday. *(limits)* Rabin (2000) — revealed today.
In passing: Sydnor (2010); Chetty (AER 2006); Rabin–Thaler (2001).
Monday: ranking whole distributions, insurance executed, and what happens when probabilities themselves are unknown — with two more reveals from your quiz.